AN2DL - First Homework Report
3Brown1Green

Davide Cagnazzo, Federica Currò Dossi, Gianluca Scanu, Davide Secco

Codabench: davidecagnazzo, fedecurrodossi, gianlucascanu, davidesecco

233217, 279987, 233946, 232579

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1 Introduction

This report aims to show the results of the Image Classification Project of Artificial Neural Networks and Deep Learning course. The aim of this project is to solve a multi classification image problem by **building a Convolution Neural Network** and exploiting the knowledge learned in class. The workflow we chose to follow relies on an **attentive analysis of the given dataset** followed by different approaches to manage class imbalances. We decided to divide the development of the model in two phases: Transfer Learning and Fine-Tuning.

2 Problem Analysis

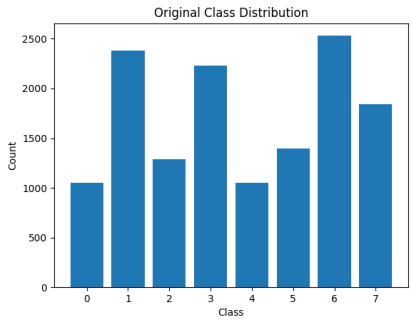


Figure 1: Distribution of the eight classes within the initial given dataset.

The original dataset is composed of 13759 images of blood cells, divided into 8 classes. In particular, the images provided are characterized by a 96 x 96-pixel size, RGB color format. The initial distribution of the 8 different classes in the dataset was as in Figure 1. The following chapters explain how we decided to tackle the main challenges of class imbalance, anomaly detection and generalization. Addressing these key issues led the way to finding the best model for our case.

3 Method

3.1 Preprocessing

Clustering has been employed in order to find anomalies and outliers within the dataset. We decided not to perform clustering directly on the images; instead, we worked with images embeddings obtained by running predictions on a Resnet50 pruned of its classification layer. Latent representation offers indeed a way to abstract pixel-level details and represent more meaningful patterns like shapes or textures. Performing K-means algorithm with progressively larger numbers of clusters allowed to us to identify groups of outliers. For ten clusters, applying PCA and t-SNE[4] with two components to obtain graphicable points, we achieved the separation showed in Figure 2:

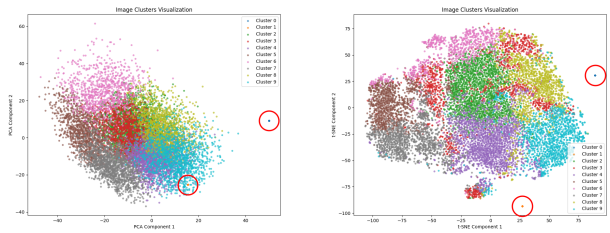


Figure 2: Outliers detection using clusters plotted using PCA and t-SNE with two components

while for three components we got the results shown in the clustering notebook. Therefore, we manually inspected samples from the clusters identifying the ones containing all anomalies to extract and delete them from the dataset. Moreover, we looked for groups of identical images with different labels, as they could cause major problems for training, but we did not find any. The new pre-processed dataset, stripped of the outlier images, now consisted of 11959 elements.

3.2 Losses

We employed the following loss functions in our experiments: **Categorical Crossentropy** and **Focal Loss**. However, Focal Loss did not improve overall performance and was excluded from the final model.

3.2.1 Optimizer

Two different optimizers were tested: Adam and AdamW. Based on our experiments, using AdamW significantly accelerated the training process. Additionally, we adopted a **CosineDecayRestarts Scheduler** to modulate the learning rate. This approach proved effective in speeding up convergence and avoiding local minima.

3.3 Managing class imbalance

3.3.1 Fixed class weights

During training, class weights were used to address dataset imbalance, but their effectiveness varied. Some well-represented classes showed low accuracy due to high intra-class variability or feature overlap, while some underrepresented classes achieved high accuracy due to their distinct features. This inconsistency made it challenging to balance training without introducing biases.

3.3.2 Dynamic class weights

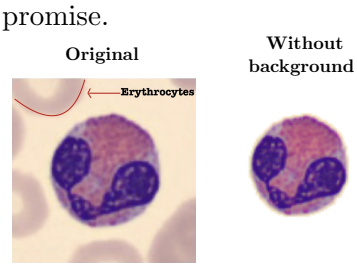
Dynamic class weights were used to address imbalance, adjusting at each epoch based on class-specific accuracy. This improved training balance, particularly for smaller networks, by focusing on underperforming classes. However, frequent updates caused instability and loss oscillations, suggesting less frequent adjustments (e.g., every 5 epochs) as a potential improvement.

3.3.3 SMOTE

In addition to the previous explained methods we tried to use SMOTE[2]. Even though the validation accuracy and loss shown improvements during Transfer Learning, after Fine Tuning the results were similar if not worse than with other methods.

3.4 Data Preprocessing: Background Removal

To help the network focus on significant cells, we removed irrelevant ones (e.g., erythrocytes) using clustering techniques and computer vision, as neural networks were not an option. It starts with a *Threshold* phase where regions are segmented based on pixel similarity. Then, during the *Dilation* phase gaps are filled in the already segmented region. Afterwards contours are refined by removing noise (*Erosion phase* and ultimately a final image is produces with segmented cells on a white background. Although not perfect, this approach showed promise.



3.5 Data Augmentation[1]

3.5.1 Offline

Employing only shifts, flips, rotations and small color variations proved to be insufficient for generalization. In the end, as shown in the augmentation notebook, we opted for stronger and more numerous transformations. Apart from techniques like CutOut, MixUp or GridMask, which were applied

at most once, we finally chose to randomly apply ten transformations to random images within each class. In the end, the final extended dataset used as input for the largest models contained around 40,000 images between the now (almost) balanced classes. We chose not to include CutMix in this work. In the context of blood cell images, it could in fact generate samples where the cropped or covered regions lack meaningful features, such as cell morphology, leading to less informative augmented data.

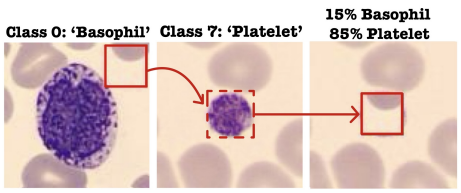


Figure 3: Problem example of CutMix application: cell disappearance.

3.5.2 Online

Stochastic data augmentation approach was designed to enhance the model's generalization capability. A module was developed to apply a predefined set of augmentation transformations, each with an associated probability. These transformations include rotations, translations, adjustments of saturation and contrast, as well as advanced techniques such as AugMix and RandomCutout.

4 Experiments

We experimented using both online and offline augmentation with different pre-trained models[3]. We started with models with fewer parameters using offline augmentation, then we tested bigger models with online augmentation. The ResNet152, a ConvNextSmall and a ConvNextLarge were tested using offline augmentation while a ConvNextSmall and a ConvNextXLarge using online augmentation. The model that used the ConvNextSmall-on doesn't have a test result since the platform didn't work.

Model	VLoss	TLoss	VAcc	TAcc	Test
ResNet152-off	0.153	0.117	0.9795	**0.9877**	0.72
ConvNeXtSmall-off	0.142	**0.114**	0.9787	0.9873	0.72
ConvNeXtLarge-off	0.128	0.181	**0.982**	0.9819	0.83
ConvNeXtSmall-on	0.226	0.386	0.9256	0.8657	–
ConvNeXtXLarge-on	**0.119**	0.318	0.9611	0.8899	**0.86**

Table 1: Results obtained. In bold we highlighted the best values

5 Results

The table 1 shows the noteworthy results and highlights a major improvement with regard to the test result by using a model with a higher number of parameters combined with online augmentation. We concluded that this was due to the fact that online augmentation allowed us to reduce overfitting by generating real time samples and virtually an infinite variety of them, which helped the model to generalize better. The adjustments made allowed our model to reach a **86% test accuracy**.

6 Discussion

The use of transfer learning and fine-tuning strategies significantly enhanced performance, particularly in addressing the high variability of blood cell images. However, challenges like effective augmentation persisted. Additionally, we could have initially focused more carefully on the choice of the model and on fine tuning parameters choice. Nevertheless, these preprocessing methods such as background removal and offline augmentation contributed to a good generalization of predictions. All considered, the model achieved competitive results, setting a foundation for future improvement.

7 Conclusions

In conclusion, this study showcased effective approaches to address a complex image classification problem, leveraging robust neural architectures and data preprocessing techniques. Looking at current limitations, several improvements can be implemented, like more advanced class balancing methods, could better handle underrepresented classes. Improving preprocessing by adopting neural-based segmentation models like U-Net or Faster R-CNN can more accurately isolate relevant cells, reducing

noise and enhancing the model's focus on meaningful features. Lastly, incorporating ensemble models that combine used architectures can improve both robustness and generalization across the dataset, leading us to higher Accuracy.

8 Efforts

While we all contributed into the whole making of the project and final model, it's important to note that we assigned ourselves different developments to execute at our bests the task that we were given. Offline data augmentation has been carried out mainly by Gianluca Scanu, along with cluster analysis. Davide Secco focused on online data augmentation and preprocessing (in particular background removal), while Davide Cagnazzo and Federica Currò Dossi contributed to the tranfer learning, fine-tuning and model design precedures. What made the difference was checking out the final work of our colleagues, to better understand the reasoning behind and further develop it.

References

[1] G. Boracchi. Artificial neural networks and deep learning, slides and materials, 2024. `https://boracchi.faculty.polimi.it/` [Accessed: 20 Nov. 2024].

[2] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. Smote: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, June 2002.

[3] F. Chollet and Contributors. Keras documentation, 2024. `https://keras.io/api/` [Accessed: 20 Nov. 2024].

[4] L. van der Maaten and G. Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(86):2579–2605, 2008.